

Chapter – 3

SAIL: An Overview

3.1 Formation and Growth of Hindustan Steel Limited (1959-1973)

When the Government of India decided to enter into the field of Iron and Steel production, it broadly envisaged not to run the firm as a departmental undertaking. Although initially steel project administration was directly under a Ministry of the Central Government, Hindustan Steel was formed as a Limited Company, with President of India owning the shares on behalf of the people of India. Thus Hindustan Steel Limited was set up on January 19, 1954.

To start with, Hindustan Steel was designed to manage with only one plant that was coming up at Rourkela. For Bhilai & Durgapur plants, the preliminary work was done by officials in Iron & Steel Ministry. From April 1957, the supervision and control of the Bhilai & Durgapur Plants were also transferred to Hindustan Steel. The registered office was originally in New Delhi, moved to Calcutta in July 1956 and ultimately shifted to Ranchi in December 1959. Initially Bokaro Project was also under HSL.

A new steel company Bokaro Steel Limited was incorporated in January 1964 to construct and operate the steel plant at Bokaro. The 1 MT phase of Bhilai & Rourkela Steel Plants were completed by end of December 1961. The 1 MT phase of Durgapur was completed in January 1962 after commissioning of wheel and axle plant. As a result, the crude steel production of HSL went up from 158 thousand tonnes (in 1959-60) to 1.6 MT (in 1961-62). 2.5 MT phase of Bhilai was completed on 2nd September, 1967 after commissioning of Wire Rod Mill. The last unit of 1.8 MT phase of Rourkela was Tandem Mill commissioned on 17th February, 1968 and 1.6 MT phase of Durgapur was completed on 6th August 1969 after commissioning of furnace in SMS. Thus, with the completion of 2.5 MT stage in Bhilai, 1.8 MT in Rourkela and 1.6 MT phase of Durgapur, the total Crude Steel output from HSL was raised to 3.7 MT in 1968-69 and 4 MT in 1972-73.

3.2 Formation of Steel Authority of India Limited (SAIL)

The Committee of Public Undertaking of the Fifth Lok Sabha was the first Parliamentary Committee to undertake a significant review of the question of setting up a Holding Company for steel. It was first considered in the Department of Steel in 1971 with the following two objectives:

- Rapid growth of the industrial sector, of the economy, of the state as a leading agent of the growth process; and
- Ability of the Government to divert investment into areas which are strategic from the point of view of future development.

In this context, it was recognized that the Public Sector had to be made more efficient in order that it might be able to contribute far more than it had to the common pool of investible surplus in the economy.

Further, such a holding company could perform a number of other important functions like coordination and control of constituent units, planning long term programmes, introduction of necessary technological changes, setting up of an R & D organisation and training of managerial personnel for the Public Sector as a whole.

Based on the above considerations, the proposal to set up a holding company for Steel and associated input industries was approved by the Government in January 1972. Accordingly, the formation of Steel Authority of India Limited was approved by the Government in December, 1972. The company was incorporated on January 24, 1973 with an authorised capital of Rs.2,000 crores.

SAIL formed in 1973, as a holding company, for :

- Hindustan Steel Ltd. (HSL)
- Bokaro Steel Limited (BSL)
- Salem Steel Limited (SSL)
- Hindustan Steel Construction Ltd. (HSCL)
- Bharat Coking Coal Ltd. (BCCL)
- National Mineral Development Corporation (NMDC)

In 1978 SAIL was restructured as an operating company :

- Steel making subsidiaries of HSL, BSL and SSL were dissolved and merged
- HSCL, BCCL and NMDC were spun off as independent companies

RINL set up in 1977 continued to be part of SAIL till it became a separate company in February'1982.

Subsequently, companies taken over as subsidiaries :

- Indian Iron & Steel Company ("IISCO") in 1979, (merged with SAIL in Feb.2006)
- Maharashtra Elektros melt Limited ("MEL") in 1986 (merged in July'2011)
- Visvesvaraya Iron & Steel Limited in 1989 (merged with SAIL in 1998)

Erstwhile Bharat Refractories Limited (BRL) merged with SAIL on 28th July'2009 and is known as SAIL Refractory Unit.

SAIL Refractory Company Ltd. (SRCL), became SAIL's subsidiary in December'2011

3.3 Present Status of SAIL

Steel Authority of India Limited (SAIL) through its five integrated steel plants at Bhilai, Bokaro, Burnpur, Durgapur and Rourkela accounts for major steel production capacity of India.

Three special steel plants at Bhadravati, Durgapur and Salem produce a wide range of special steels, special alloy steels and stainless steel.

Chandrapur Ferro Alloy Plant is the only Public Sector Unit engaged in production of Manganese based Ferro Alloys in the Country

Today, SAIL is one of the largest corporate entities. Its innate strength lies in its technologists and professionals and a trained manpower of 50,246 as on 01.02.2026. During 24-25, the company has clocked its best-ever sales of 18.07 MT while achieving revenue from operation of Rs.1,02,478 Cr.

3.4 Expansion & Modernisation of SAIL:

In order to maintain market competitiveness and to meet the challenging needs of its customers, SAIL is undertaking capital investment projects from time-to-time. This is being done by way of Technological Upgradation, Addition/ Modification/ Replacement of its capital assets, overcoming of production bottlenecks, improving the productivity of various units, quality of products, improving health of plant & equipment, improving yield, conserving energy, effecting economy in the cost of production & better techno-economics and abating environmental pollution. Presently, SAIL is also increasing its focus on undertaking projects for reduction of Carbon Footprint.

SAIL is presently having Crude Steel operating capacity of 20.30 MTPA through its five Integrated Steel Plants located at Bhilai, Bokaro, Durgapur, Rourkela and Burnpur. Apart from these five Integrated Steel Plants, at present, SAIL has two Special Steel Plants located at Durgapur and Salem having Crude Steel operating capacity of 0.41 MTPA.

Increase in the Crude Steel capacity to around 50 MTPA is envisaged through augmentation of its existing facilities as well as through Brownfield/Greenfield expansion of its Integrated Steel Plants at Bhilai Steel Plant, IISCO Steel Plant, Rourkela steel Plant, Durgapur Steel Plant and Bokaro Steel Plant.

However, the expansion is subject to following:-

- (i) Demand Growth for finished steel to absorb the enhanced level of steel production in the sector.
- (ii) Resource availability to finance CAPEX with sustainable Debt: Equity ratio.
- (iii) Captive iron ore sources for supporting the enhanced capacity.
- (iv) Availability of Land

Increase in the Crude Steel Capacity of Integrated Steel Plants of SAIL is envisaged as under:-

Steel Plant	Crude Steel Capacity (MTPA) - Current	Envisaged Crude Steel Capacity (MTPA)(FY 2047)
Bhilai Steel Plant (BSP)	6.80	12.00
Durgapur Steel Plant (DSP)	2.20	7.50
Rourkela Steel Plant (RSP)	4.20	8.80
Bokaro Steel Plant (BSL)	4.60	14.0

IISCO Steel Plant (ISP)	2.50	7.30
SAIL (5 ISPs)*	20.30	49.60

* The capacities of Special Steel Plants of SAIL have not been included in these figures.

The implementation of the expansion projects at different Plants are proposed to be taken up in a phased manner with a view to keep the CAPEX and the borrowings within manageable limits. Various alternatives are being explored to optimize the investments for capacity enhancement in each Plant. Focus is also on undertaking projects for reduction of Carbon Footprint.

3.5 Different Plants & Units of SAIL

Bhilai Steel Plant (BSP)

An agreement was signed in New Delhi on February 2, 1955 between the Government of India and Soviet Union to set up an integrated steel plant at Bhilai with a capacity of 1 MT of ingot steel. The plant began its operation on January 31, 1959 when Coke Battery No. 1 was commissioned. Production of Pig Iron at Bhilai began on February 4, 1959 when Blast Furnace No. 1 was commissioned.

Situated in Chhattisgarh, this was one of the three 1 MTpa capacity crude steel plants set up in the Public Sector in the late fifties. Subsequently it was expanded to 2.5 MT ingot capacity.

The plant was the first in India to produce wide (3600 mm wide) heavy plates. A major exporter of steel products, Bhilai specialises in shaped products, such as heavy rails, heavy structurals, merchant products and wire rods.

Eleven times winner of Prime Minister's Trophy for Best Integrated Steel Plant in the country, Bhilai Steel Plant (BSP) is India's largest producer & supplier of world class rails for Indian Railways including world's longest 130 metre rails in single piece and 260 metre long rail welded panels, and a major producer of large variety of wide and heavy steel plates and structural steel. The plant also specializes in other products such as wire rods and merchant products. The entire range of TMT products (Bars & Rods) produced by the Plant is of earthquake-resistant grade and superior quality. The plant also produces heavy structurals including channels and beams.

Current production capacity of the plant is as under :

Hot Metal – 7.2 MT

Crude Steel – 6.8 MT

Saleable Steel – 6.4 MT

Product Mix:



Product Mix
Semis
Rails & Structural
Merchant products
Wire & Bar rods
Plates

Rourkela Steel Plant (RSP)

RSP was the first of the three steel plants taken up in the Public Sector. On December 31, 1953, an agreement was made between the Government of India and a Consortium consisting of Thyssen & Demag, Aktiengesellschaft, Duisburg, Germany to set up a steel plant of initial capacity of 0.5 MT subsequently a supplementary agreement was signed in July 1955 to set up a 1 MT plant. The Coke Oven Battery No.1 was commissioned on 3rd December, 1958 and the first of the three Blast Furnaces was commissioned on 3rd February, 1959. A major producer of diversified range of sophisticated steel products, RSP is among India's few Plants producing 100% of the steel through the globally profuse Continuous Casting route since 1998. RSP is the only SAIL plant having capability to produce large dia pipes.

Situated in Odisha, it was the first integrated steel plant in India to produce flat products and the first in Asia to introduce basic oxygen furnace (BOF) process at a time when this process was yet to receive recognition from the established steel producers at home and abroad. The plant produces a wide range of flat steel products like plates, hot and cold rolled coils and sheets, galvanized sheets, electrical steel sheets and large diameter electric resistance welded (ERW) and spiral welded (SW) pipes. The plant was expanded in the late sixties (1965-68) from 1.0 MT to 1.8 MT per annum ingot steel capacity. Continuous technological innovation has led to greater diversification in this plant's product range. It also produces cold rolled non-grain oriented (CRNO) sheets. RSP is geared up producing defence and space quality plates through a Special Plate Plant.

The Plant has undergone modernization upgradation in two phase. The Phase-I of modernization completed in 1994 and Phase-II modernization completed in 1997-98. After modernization, the capacity got augmented to 2 MTs of hot metal and 1.9 MTs of crude steel.

Current production capacity of the plant is as under :

Hot Metal – 4.65 MT (Million Tonnes)

Crude Steel – 4.2 MT

Saleable Steel – 4.3 MT

Product Mix:



Flat

PRODUCT-MIX RSP
Plate Mill Plates
HR Plates
HR Coils
ERW Pipes
SW Pipes
CR Sheets & Coils
Galvanized Sheets (GP& GC)
CRNO steel

Durgapur Steel Plant (DSP)

Situated at a distance of 158 km from Calcutta, the Grand Trunk Road and the main Calcutta-Delhi railway line pass through Durgapur. It is situated on the banks of the Damodar river.

Durgapur Steel Plant (DSP) set up in late fifties, is a leading producer of long products and is the pioneer in manufacturing and supply of forged Railway Wheels & Axles in the country. DSP started production with an initial crude steel capacity of 1 MPTA in 1959, which has been progressively increased to 1.8 MTPA during the modernization in nineties and further to 2.2 MTPA during recently completed Modernization & Expansion Plan (MEP).

Current production capacity of the plant is as under :

Hot Metal – 2.40 MT (Million Tonnes)

Crude Steel – 2.20 MT

Saleable Steel – 2.12 MT

Unique feature of this plant is its Wheel & Axle Plant for making forged wheels and axles, catering to Indian Railways. Over the years plant has developed various types of wheels as per need of Railways.

Products from DSP's state of the art Medium Structural Mill (MSM) have been accredited with stringent CE marking certificate to promote their export.

Product Mix:



Product-mix
Merchant products
Structural
Wheels and Axles
Semis

Bokaro Steel Plant (BSL)

Bokaro Steel Plant 'brings out before one's eyes the vision of a massive giant in the making'. As the fourth steel plant in the Public Sector, conceived in 1959, it actually started taking shape in 1965 with the signing of an agreement with the Government of USSR on 25th January 1965. Envisaging a capacity of 1.7 MT in 1st stage and 4.0 MT in 2nd stage, its construction started on 6th April, 1968.

Bokaro Steel Plant was originally incorporated as a Limited Company on 29th January 1964. After the formation of SAIL in 1973, it became a wholly owned subsidiary of SAIL and on 1st May 1978 it was eventually merged with SAIL in terms of Public Sector Iron & Steel Companies (restructuring) and Miscellaneous Provisions Act 1978.

The plant was conceived as the country's first 'Swadeshi' steel plant, to be built with maximum indigenisation going into the equipments, materials and know-how. Thus, this project has been a radical shift from the earlier dependence on foreign sources of know-how and consultancy, design and equipment, engineering, supervision and erection to almost a full measure of self-reliance and confidence.

The Plant is hailed as the country's first Swadeshi Steel Plant, built with maximum indigenous content in terms of equipment, material and know-how. Its first Blast Furnace started on 2nd October 1972 and the first phase of 1.7 MT Ingot steel was completed on 26th February 1978 with the commissioning of the third Blast Furnace. All units of 4 MT stage have already been commissioned and the modernisation taken subsequently has further upgraded this to 4.65 MT of liquid steel.

Bokaro is designed to produce flat products like Hot Rolled Coils, Hot Rolled Plates, Hot Rolled Sheets, Cold Rolled Coils, Cold Rolled Sheets and Galvanised Plain Sheets/Coils.

Bokaro is geared to provide a sure and strong raw material base for a host of modern engineering industries like motor vehicles, pipes and tubes, cold rolling units, barrel and drum making and lately, LPG cylinders. Galvanized plain and corrugated sheets are finding use in industrial and domestic applications.

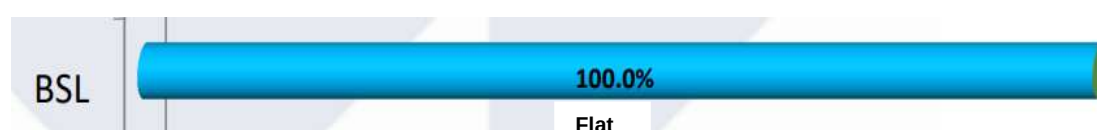
Current production capacity of the plant is as under :

Hot Metal – 5.25 MT (Million Tonnes)

Crude Steel – 4.65 MT

Saleable Steel – 4.24 MT

Product Mix:



Product-mix
HR Coils, HR Plates, HR sheets
CR Coils and Sheets
GPsheets/coils

IISCO Steel Plant (ISP), Burnpur

This full-fledged integrated steel plant is one of India's oldest. Situated at a distance of about 200 kms from Kolkata in Burnpur on the banks of the perennial Damodar river in West Bengal, ISP is well connected by both South Eastern and Eastern Railways and National Highway 2. Its proximity to Kolkata and Haldia ports is an additional strength.

Promoted by Burn & Co., the Indian Iron and Steel Company was incorporated on 11 March 1918. The iron works were set up at Hirapur. IISCO produced iron from an open-top blast furnace at Hirapur in West Bengal.

An open-top blast furnace, had earlier been set up in 1870 at Kulti, by a company known as Bengal Iron Works Co. (BIW), founded by James Erskine. The plant was taken over by the Government in 1881. In 1889, it was taken over by the newly formed Bengal Iron & Steel Co. Around 1904, Bengal Iron & Steel Co. (BISCO) converted the open top furnaces into closed top furnaces, and installed facilities for producing steel, through open hearth furnaces. The unit held the proud distinction of being the owner of India's "oldest unit producing pig iron by modern methods" at Kulti.

BISCO was absorbed by IISCO in 1936 and steel making started as a regular measure in 1939.

Another company named Steel Corporation of Bengal (SCOB), incorporated in 1937, was also amalgamated with IISCO in 1952. SCOB's Napuria Works and IISCO's Hirapur Works in unison came to be known as the Burnpur Works of IISCO.

The Burnpur Works underwent two overlapping expansion in 1953 and 1955, increasing its production capacity to 1 MT of ingot steel and 0.8 MT of saleable steel.

IISCO had acquired iron ore mines at Gua and Chiria in what is today's Jharkhand state and collieries in Chasnalla and Jitpur (also in Jharkhand) and Ramnagore (in Bengal). These captive sources of high quality raw materials gave IISCO a major competitive edge and enabled it to establish a prestigious reputation in domestic and foreign markets. It also became the first Indian blue chip company to have its shares traded at the London Stock Exchange.

Despite growth plans, however, a combination of factors drove the company into stagnation and decline, resulting in lack of investment for technology upgradation to meet emerging market competition. IISCO was nationalised in 1972 and became a wholly-owned subsidiary of SAIL in 1979. The Indian Iron & Steel Company (IISCO), a SAIL subsidiary, was amalgamated with SAIL on 16th February 2006 and renamed IISCO Steel Plant (ISP).

After undergoing a modernisation-cum-expansion programme, the crude steel capacity of the plant has been raised to 2.5 million tonnes per year. The facilities have been designed to produce best quality product with minimal environmental impact.

The plant manufactures a range of products, over some of which it holds exclusive market dominance. Iron & steel produced by it has been acknowledged as being of the finest quality.

Current production capacity of the plant is as under :

Hot Metal – 2.7 MT (Million Tonnes)

Crude Steel – 2.50 MT

Saleable Steel – 2.39 MT

Product Mix:



Product-mix
Wire Rod, TMT Re-bar
Structurals
Semis for sale

Alloy Steels Plant (ASP)

A pioneer in the production of alloy and special steels, Alloy Steels Plant (ASP), Durgapur was commissioned with an initial capacity of 1,00,000 tonnes of ingot steel and 60,000 tonnes of saleable steel. Through two phases of expansion and modernisation, the capacity has been revised to 2.46 lakh tonnes of liquid steel and 1.78 lakh tonnes of saleable steel.

ASP is equipped with state-of-the-art technology for producing world class quality alloy and special steels. The plant has one slab-cum-twin bloom continuous casting machine, the only one of its kind in India. It is specially designed for casting special steels like Austenitic and Ferritic stainless steel and a variety of non-stainless steels including bullet-proof steel. ASP has the capacity to produce Slabs, Blooms, Bars, Plates and Forged items of over 400 grades in a wide range of sizes.

Salem Steel Plant (SSP)

Salem Steel Plant (SSP), is a premier producer of international quality stainless steel in India. Commissioned in 1981, Salem Steel Plant, a special steels unit of Steel Authority of India Ltd, pioneered the supply of wider width stainless steel sheets / coils in India. The plant can produce austenitic, ferritic, martensitic & low-nickel stainless steel in the form of coils & sheets with an installed capacity of 70,000 tonnes / year in Cold Rolling Mill & 3,64,000 tonnes / year in Hot Rolling Mill. Its steel melting shop can produce 1,80,000 tonnes of slabs per annum. In addition, the plant has country's first top-of-the-line stainless steel blanking facility with a capacity of 3,600 tonnes / year of coin blanks & utility blanks / circles.

Under conversion scheme, value-added products like kitchen & tableware and doorframes are manufactured and supplied in bulk to corporates. SSP has also developed new applications of its products viz. LPG tanks for automobiles, stainless steel ceiling fans, exhaust fans, corrugated sheets, water tanks, etc.

Visvesvaraya Iron & Steel Plant (VISP)

Visvesvaraya Iron and Steel Plant (VISL) is a pioneer in production of high quality alloy & special steels and pig iron. Steel is produced through BF-BOF-LRF-VD route. The facilities include vacuum degassing, vacuum oxygen decarburisation, ladle refining furnaces, ingot teeming, continuous casting, 1600 Tonnes-hydraulic-high-speed forging press, a fully automatic horizontal long forging machine with high programmable Logic Controller system for a semi-automatic and automatic mode of operation. VISL has an installed capacity of 2,16,000 tonnes of hot metal and 98,280 tonnes of alloy & special steels.

Chandrapur Ferro Alloy Plant (CFP)

Chandrapur Ferro Alloy Plant, (CFP) erstwhile Maharashtra Elektros melt Ltd. (MEL) has become a Unit of SAIL w.e.f. 12/7/2011. Chandrapur Ferro Alloy Plant is the only Public Sector Unit engaged in production of Manganese based Ferro Alloys in the Country.

CFP has an installed capacity of 1,24,000 TPY Silico-Manganese. The product range of CFP includes High Carbon Ferro Manganese, Silico Manganese and Medium/Low Carbon Ferro Manganese. The Plant is accredited with Quality Assurance Certificate ISO 9001:2008. CFP's major production facilities include two nos. of 33 MVA and one no. 45 MVA Submerged Electric Arc Furnaces, two nos. Manganese Ore Sintering Plants, Furnace gas based Power Plant and 1 MVA Electric Arc Furnace for the production of MC/LC Ferro Manganese with Lime Calcination and Manganese Ore Roasting Unit.

SAIL Refractory Unit (SRU)

The erstwhile Bharat Refractory Limited (BRL) was merged with SAIL with effect from April 2007. Following merger it has now been renamed as SAIL Refractory Unit (SRU). It caters to the refractory needs of all SAIL Plants. Main objectives of SRU are:

SRU has four production units, out of which three are situated in Jharkhand and one in Chhattisgarh. The units at Jharkhand include (i) Bhandaridah with a production capacity of 26,000 tons of Tap Hole Mass for Blast Furnace, various Castables and Fire Clay Bricks, (ii) Ranchi Road with a production capacity of 7,500 tons of Basic Masses and Magnesite carbon Bricks and (iii) IFICO, Ramgarh with a production capacity of 42,000 tons of Slide Gate Plate, different types of Castables and High Alumina Bricks. The unit at Chhattisgarh is situated in Bhilai is engaged in producing the entire range of basic and silica refractories. It also carries out calcination of lime in its high capacity rotary kiln, which is further used by Bhilai Steel Plant for iron & steel production.

Captive Mines

SAIL has the second largest mining outfit in the country after Coal India Ltd. Spread over the mineral rich states of Jharkhand, Odisha and Madhya Pradesh, the mines of SAIL started their operations as captive sources of raw materials of its integrated steel plants. By virtue of their locations and also having developed under the different steel plants for more than 2 to 4 decades, they present a picture of fascinating diversity, not only in the nature of their reserves/deposits but in their legacies as well, with each one of them being remarkably distinct from the other.

1) Iron ore Mines

SAIL operates 15 iron ore mines spread over the states of Jharkhand, Odisha and Chhattisgarh. There are 22 mining leases held by SAIL for operation of these mines. Most of these leases were granted in the 50's and 60's or earlier and almost all of them are undergoing second/subsequent renewal/extension of lease along with various statutory clearances.

	JHARKHAND	ODISHA	CHHATTISGARH
Mine	Kiriburu, Meghahatuburu, Gua and Chiria	Bolani, Barsua & Kalta, Taldih	Rajhara Mechanical, Mahamaya, Dulki, Jharandalli, Dalli Mechanical, Rowghat and Kalwar
Lease No	11	5	6

Barring a few most of these leases have been extended as per Mines & Minerals (Development & Regulation) (Amendment) Act, 2015.

Iron ore Production :

Iron ore mines of SAIL currently produce about 34 Million Tonne Per Annum (MTPA) of iron ore. Details of the iron ore produced during last five years including current year are placed below:

Quantity: Million Tonnes

MINE	2018-19	2019-20	2020-21	2021-22	2022-23	2023-24	2024-25
JHARKHAND							
Kiriburu	3.89	3.91	3.83	4.05	4.05	3.62	3.81
Meghahatuburu	3.66	3.73	2.97	3.56	3.67	3.24	2.5
Gua	3.67	3.69	3.10	4.15	3.96	4.16	4.02
Chiria (Manoharpur)	0.73	0.74	0.55	0.08	0.34	0.49	0.41
ODISHA							
Bolani	5.63	5.92	6.24	7.05	7.11	7.2	7.41
Barsua	1.86	1.75	2.26	2.55	2.29	2.48	1.85
Taldih	0.30	1.01	1.09	1.25	1.34	1.34	1.76
Kalta	1.72	1.71	2.49	3.17	3.19	3.19	3.37
CHHATTISGARH							
Rajhara Mechanical	3.52	3.46	3.98	3.49	3.54	3.68	6.93
Mahamaya							
Dulki							
Jharandalli	3.37	3.36	3.55	4.80	4.22	4.37	
Dalli Mechanical							
Rowghat	-	-	-	-	0.07	0.46	10.01
Kalwar						0.02	0.71
TOTAL	28.35	29.28	30.06	34.15	33.78	34.34	33.78

Iron Ore Mines Expansion: Aligned with the objectives of the National Steel Policy (NSP) 2017, which envisages achieving 300 MTPA steel capacity by FY 2030-31, SAIL has drawn up a phased plan to enhance its hot metal capacity from about 20 MT in FY 2024-25 to nearly 37 MT. Achieving this growth hinges on long-term raw

material security, particularly iron ore, and increased adoption of beneficiation and pelletisation for low-grade ores. Accordingly, SAIL is strengthening its mining capacity along with beneficiation, crushing, pellet plants, and logistics infrastructure. Against an installed iron ore capacity of 46 MTPA, production stood at around 34 MTPA in FY 2024–25 to support crude steel output of 19.17 MT. To meet future requirements, the company plans to scale up iron ore production capacity to about 100 MT by FY 2030-31.

1.2 Coal Mines

Coking Coal is critical raw material required by the steel industry. Indigenous availability of appropriate qualities of coking coal is limited due to which the steel industry is dependent upon imported coking coal. SAIL imports about 80-85% imported coal and blends it with coking coal from own washery and washed coking Coal available from CIL.

Coking coal is a critical input for steelmaking, and due to limited domestic availability, SAIL meets about 80-85% of its requirement through imports, blending it with coal from its own mines and sourced from BCCL/CCL (Coal India Limited).

SAIL operates three coal mines:

- Chasnalla & Tasra (Jharia, Jharkhand) – produce coking coal
- Ramnagore (Raniganj, WB) – produces non-coking (thermal) coal

Raw coking coal from Chasnalla, Tasra and Indigenous Sources (i.e. BBCL) is washed at Chasnalla Washery, which produces about 0.4 MTPA washed coking coal (18% ash) for SAIL steel plants during FY 24-25. Middlings were also produced which is used for power generation. Washed coking coal is also sourced from BCCL/CCL washeries. Thermal coal from Ramnagore is supplied to steel plants through Chasnalla siding after blending with middlings.

Moreover, SAIL is developing the Tasra Coal Block of 4 MTPA and a 3.5 MTPA Coking Coal Washery through the MDO.

Production figures of Collieries for last 3 years

(Unit: '000 Tonnes)

Financial Year	2022-23	2023-24	2024-25
RAW COKING COAL			
Chasnalla	61	104	86
Jitpur	8	27	4
Tasra	303	334	498
Total Raw Coking Coal	372	465	588
CLEAN COKING COAL			
Chasnalla Washery	619	484	376
THERMAL COAL(Including Jhama& Middlings)			
Ramnagore Incl. Jhama	16	75	98
Middlings	553	644	737

Financial Year	2022-23	2023-24	2024-25
RAW COKING COAL			
Total Thermal Coal	569	719	835

1.3 Limestone, Dolomite mines : In addition, SAIL holds captive mining leases of limestone and dolomite mines -

1. Kuteshwar limestone Mines, M.P.
2. Nandini Limestone Mine, Chhattisgarh
3. Hirri Dolomite Mine, Chhattisgarh

Production details of operational flux mines are provided below:

Quantity: Million Tonnes

Sl	Mines	2022-23	2024-25	2025-26 (Apr'25-Dec'25)
1	Kuteshwar Limestone	1.06	1.10	0.50
2	Nandini Limestone	0.30	0.21	0.15
LIMESTONE TOTAL		1.36	1.31	0.65
1	Hirri Dolomite	0.45	0.40	0.44
DOLOMITE TOTAL		0.45	0.40	0.44
SAIL FLUX		1.81	1.71	1.09

Central Coal Supply Organisation (CCSO)

CCSO, Dhanbad, is responsible for daily movement of washed Coking Coal and Power grade coal to all Integrated Steel Plants of SAIL and also coordinates movement of Raw Coal from BCCL to Chasnala Washery.

Functions of CCSO

- 1) Linkage planning for all steel Plants under SAIL.
- 2) Proper assessment of quality of indigenous coal at loading points.
- 3) Liasioning with the Coal Companies to supply coal as per the desired quality and quantity to SAIL Plants.
- 4) Liasioning with Railways to supply rake as per the requirement plan of SAIL Plants.
- 5) Liasioning with third party for joint sampling of rakes at loading point.
- 6) Maintaining all records and statistics related to production and dispatch of coal.
- 7) Exploring new source of coking coal for washing at Chasnala and other Washeries.
- 8) Ensuring long terms security of Coking Coal for SAIL Plants through participation in linkage auction of Coking Coal conducted by CIL.

- 9) Meeting short term additional requirement of Coking Coal through participation in exclusive auction of Coking Coal conducted by CIL.

Supply of Coal from CCSO

SAIL / CCSO currently plans to supply approx. 2.6 MTPA of washed coking coal and 4 MTPA of Boiler coal from domestic sources including captive colliery at Chasnalla to different SAIL Plants.

Central Marketing Organisation (CMO) : -

SAIL's marketing set-up, is the India's largest industrial marketing set-up. The Sales & Marketing of all prime products produced by SAIL is undertaken by Central Marketing Organisation (CMO), except for Stainless Steel produced by Salem Steel Plant (SSP) which is directly dealt by SSP. However, Mild Steel Hot Rolled Coils produced at Salem Steel Plant are sold through CMO.

Steel produced at SAIL Plants is marketed by **Central Marketing Organisation (CMO)** Hq at Kolkata through its extensive network of-

- ✓ 4 Regional Offices (*Delhi, Kolkata, Mumbai and Chennai*) and International Trade Division (ITD) at Kolkata
- ✓ 35 Branch Sales Offices
- ✓ 4 Customer Contact Offices
- ✓ Retail network Set Up Tier 1 and Tier 2

Serviced through-

- ✓ 35 Warehouses
- ✓ Direct dispatch from Steel Plants
- ✓ Door Delivery
- ✓ Retail Channel

In addition to sale of steel products, other activities performed by CMO include Warehousing & Logistics, Branding & Brand promotion, Market Analysis and Research, Development of new products, Creation and Management of new Distribution Channels etc.

SAIL exports steel products through its International Trade Division (ITD), Kolkata.

Total Sales during FY 2024-25 was about 18.07 MT, including 0.12 MT in exports.

SAIL has been contributing in Nation building since inception by servicing the requirement of various Infrastructure Projects and also projects of Strategic Importance. The focus on Nation building has expressed itself in the growth of sales in different categories like TMT, Structurals and PM Plates.

The CMO Head Quarter is located at Kolkata.

Wide range of Products:



PIG IRON



COLD ROLLED PRODUCTS



PIPES



SEMIS



STRUCTURAL



TMT



GALVANISED PRODUCTS



BARs, RODS AND REBARS



PLATES



RAIL



WHEELS AND AXLES



HOT ROLLED PRODUCTS



STAINLESS STEEL PLATE



ELECTRICAL STEELS



SAIL SEQR TMT BARS



SAIL UTENSILS



ASP SLAB



ASP ROLLED ROUNDS



GP/GC SHEET



SAIL NEX- STRUTURALS

The products handled by CMO include Hot Rolled Coils & Sheets, Cold Rolled Coils & Sheets, Galvanised Products, Plates, Reinforcement Bars, Structural, Rails, Wheels & Axles, Wire rods, Pipes & Tubes, Electrical Sheet, Semi-finished Steels, Alloy Steel products, etc. in numerous grades and sizes.

Inspired by the Government's clarion call for "Atmanirbhar Bharat" and "Vocal for Local", SAIL is relentlessly making efforts in developing grades/sizes, which can play important role in these Missions. The new grade of steel developed by SAIL's Salem Steel Plant "Duplex Stainless Steel" is an effort in that direction. This steel has superior corrosion resistance with higher strength and formability and has been mainly imported so far.

SAIL came up with new grades of steel like -

- **ISP:** AWS A5.17 EM12K, 5.5 mm and AWS A5.23 EA2 (Mo bearing) at WRM for welding applications
- **DSP:** Ultra Low Nb Structural (E-350 BR) IS 2062 E410 C. Channel 300 was rolled at DSP for first time.
- **BSP:** IS 7887 CAQ Gr1 and EWNr Wire Rods 5.5/6/7mm at WRM
- **RSP:** HSFQ 450 and MC-30/40/55 at HSM II
- **BSL:** EN 10025-5/ IS 11587 HR Coils at HSM.

International presence:

SAIL has opened Representative Office in UAE, Mozambique & Nepal to liaison with customers & suppliers.

International Trade Division of CMO, located at Kolkata, is responsible for exports of SAIL products. During 2024-25, exports contributed to 0.48% (492 Rs. Cr.) of the total turnover

SAIL is focusing on geographically closer markets (SAARC nations and Europe) and diversifying its export basket with products from multiple Plants, including BSL, ISP, DSP, and RSP.

SAIL has a leadership position in consumer mind space and is always striving to bridge the ever changing expectation gaps with improved services, products and processes.

Some of the Marketing efforts and initiatives by SAIL are broadly listed below:

Development of new products and customer outreach

Continuous efforts are being made at SAIL to develop new products required by steel users. New products have been developed for various commercial and strategic applications like Steel for Shipping containers, Weather resistant steel for Vande Bharat Coaches, High Strength Steel for Pre-Engineered Buildings, Fire-resistant steel, Seismic grade TMT bars, Special steel plates for submarines, New Electrical steel grades, High Tensile Structurals & Plates with improved ductility for Defence, Various grades of API for Oil & Gas pipeline; High Tensile grades for EME segment, High Carbon Wire rods for Wire drawing, etc. SAIL supplied major portion of Stainless steel to Railways- Vande Bharat Coaches.

SAIL has continued the thrust on "NEX " brand of structurals. Efforts have been made to popularise usage of steel in designing in general and increase usage of "NEX " Brand of SAIL Structurals through virtual meeting with customers, webinars and participated in virtual workshop sessions in-house and also by structural designers & Architects. To popularize and give impetus to the use of SAIL Structurals, company has undertaken various campaigns to promote NEX from SAIL. Further various creative designs and content have been developed which is being used in form of posters , banners , standee and hoardings during various customer meets/ Architect & Structural Designer meets to popularize NEX Brand of SAIL Structurals. In addition Tier I Distributors of Structurals are promoting NEX brand through advertisements/hoardings etc.

SAIL has an extensive dealership network comprising more than 4800 dealers spread across the Country. SAIL is in the process of strengthening its retail channel to reach steel to the hinterland and individual steel consumers. In order to reach out to the end customer in the Retail Segment through an efficient distribution channel and providing value addition through products, delivery and services to customers, a 2-Tier Distributor Channel has been established by SAIL across the Country.

For enhancing usage of steel in designs, SAIL has been conducting workshops/lectures at engineering colleges across the Country. Seminars have been held for architects and designers on the theme “New Challenges in Steel Design & Construction” for promoting steel structure designs.

Cross functional teams comprising representatives from Marketing and Plants are formed for development of new products. Physical meetings/interactions over virtual modes are organized with various stakeholders to enhance product knowledge, facilitate product development and create product recognition and recall.

SAIL has been actively pursuing with various Ministries to put emphasis on Life Cycle Cost in projects and on Government initiatives aiding increased usage of Steel.

SAIL participates in webinars, conferences, seminars, etc. on Steel and its' usage conducted by Industry & professional associations and institutes, including sponsoring programs where deemed suitable.

SAIL is actively utilizing its Social Media accounts like Twitter, Facebook, Instagram, LinkedIn, YouTube etc. for promoting Steel Usage.

Important customers are provided single window service by a dedicated Key Account Manager. This has improved communication, quality of service and customer satisfaction, particularly for large customers who are spread over multiple locations across the Country and would otherwise need to contact SAIL at many points.

Efforts to enhance Availability and Reach

Towards digitalization AI based Chatbot ‘**SAIL SARATHI**’ has been introduced for facilitating easier navigation & information accessibility for customers and visitors.

SAIL also owns an E-portal (www.sailsuraksha.com) where customers can book TMT orders online.

In order to help promote and build acceptance of steel usage in rural areas, SAIL has an ongoing programme under “Gaon Ki Ore” campaign, for working with village level public decision makers, opinion makers, masons, builders, etc. Under the campaign, 363 workshops have been conducted during the financial year 2022-23 across the Country with focus on small consumers, etc. During Apr-May '23-24, 90 workshops have been held.

Efforts to Improve Sales

- Key Accounts Management to develop strategic relationship with high value customers.(62 Numbers)
- Tier-1 Distributorship scheme to increase customer outreach & Tier-2 Distributorship scheme to offer customized products through retail channel .
- Contract Manufacturing with M/s NMDC Steel from Dec 2024 onwards to add more volume sales of HR products.
- SAIL has inducted TMOs (Technical Marketing Officers) to Solve technical problems of customers in usage in field, evolving needs & identify new areas / new applications for usage of SAIL Steel .
- Seminars have been held for architects and designers on the theme “New Challenges in Steel Design & Construction” for promoting steel structure designs. Interacting with the Government Agencies, Central PSUs, Private Construction Companies, Municipal Authorities etc. for propagating usage of steel. Senior executives have been participating in industry specific seminars and webinars like ones on Railway Containers, Defence, Railways, Highway, Housing, PEB Segment, Oil & Gas Sector and potential in rural economy, etc.
- Organized Defence Conclave 2025 to foster deeper collaboration with Defence Public Sector Undertakings /Ordinance factories.
- Supplying critical naval grades to defence :
 - India’s first indigenously designed and constructed Anti-Submarine Warfare Shallow Water Craft INS Arnala
 - Prestigious vessels such as INS Vikrant, INS Vindygiri, INS Nilgiri, INS Udaygiri , INS Himgiri and INS Surat among others.
 - Two significant naval platforms INS Ajay and INS Nistar etc.
- Digitalization initiatives taken by SAIL to ensure ease of business:
 - Inclusion of TMT Estimator in SAIL Suraksha Web site – With the help of INSDAG, TMT estimator has been provided on SAIL-Suraksha web site for the estimate of size wise TMT required for building. e-portal <https://sailsuraksha.com/>
 - Integration with SBI for payments from customer receipts done.
 - MSME Registration Module in SAIL Portal.
- Developed wheels for the Vande Bharat Express from its Wheel & Axle Plant,Durgapur.
- To meet Indian Railways requirement for higher axle load, SAIL BSP has developed & supplied R 350 HT grade Long Rail Panels. This product has been made in India for the first time.

- Based on changing market requirement, SAIL is continuously engaged in product improvement and product development to keep pace with the demand scenario for various end use segments .

Research & Development Centre for Iron & Steel (RDCIS)

Research & Development Centre for Iron and Steel (RDCIS) is the corporate R&D unit of SAIL. Set up in 1972, RDCIS is registered under DSIR as an approved R&D unit. RDCIS is the nodal agency for product development and intellectual property management in SAIL. It undertakes R&D projects in diverse realms of iron and steel technology under the categories of Basic Research, Product Development, Plant Performance Improvement, Scientific Investigation and Development and Technical Services. RDCIS has around 121 dedicated and competent scientists and engineers who are specialists in their fields.

RDCIS provides customers with prompt, innovative and cost-effective R&D solutions, develops and stabilizes new products and continually enhances the capability of its human resources to emerge as a centre of excellence. The major efforts are directed towards product development, cost reduction, quality improvement, energy conservation and value-addition to products of SAIL and in providing application engineering support for products at customers' end through a well established marketing network of SAIL.

RDCIS offers technological services to various organizations in the form of transfer of know-how of technologies developed by RDCIS, consultancy services / contract research, specialized testing services and training. This helps to establish its credibility as a knowledge centre, besides generating revenue through external earning.

RDCIS also enters into collaborative agreements with various institutes of national repute, Govt. of India organizations and international organizations to upgrade its knowledge base and develop new technologies and products.

RDCIS plays an increasingly vital role towards meeting the current as well as future needs of the SAIL steel plants/ units and its external customers. This depends on the availability of research facilities and their utilization. The Centre has created state-of-art facilities to ensure in-depth scientific research in diverse fields of iron and steel technology. It is equipped with more than 500 advanced diagnostic equipment and 5 pilot facilities under 15 major laboratories.

SAIL Safety Organisation (SSO)

SAIL Safety Organization (SSO), a Corporate Unit set up in 1988 at Ranchi, monitors and guides the safety Promotional, fire and Occupational Health Services activities undertaken at different steel Plants/Units/Mines/Stockyards. To accomplish the above mentioned functions, SSO formulates and prepares appropriate safety policies, procedures, systems, action plans, guidelines etc. and follows up for their implementation and thereby helps in providing accident free work environment. Consistent efforts are also being made by SSO for competence building in the area of

safety management through HRD interventions covering heads of shops, line managers, safety personnel & trade union leaders.

A multi-disciplinary safety Engineering Departments exists in each of the steel plants and mines to look after their safety needs. The emphasis is now on Systematic Approach to safety Management. SSO is managing the secretariat of the Joint Committee on Safety, Health & Environment in the steel Industry (JCSSI), a bipartite forum which addresses steel plant safety, health & environment issues with active involvement of major central trade unions and management of major steel producers of the Country, on promoting safety. Occupational health and pollution control measures.

Centre for Engineering & Technology (CET)

Centre for Engineering & Technology (CET), is the design, engineering & consultancy unit of SAIL which was started in 1982. It has its Head Office at Ranchi, Sub Centres at Bhilai, Durgapur, Rourkela, Bokaro and an IPSS Secretariat at New Delhi for formulation of Interplant Standards for Steel Industry. As a 'solution provider for all project needs', CET has been rendering complete range of services not only to the steel plants under SAIL but also to various clients other than SAIL – both within and outside the country. CET is also the nodal agency for acquisition and lateral transfer of technologies within SAIL plants.

The range of services includes conceptualisation, project evaluation & appraisal, project consultancy, design & engineering and project management in the areas of iron and steel making. Apart from this, CET has been providing its services in the related areas like mine planning and development, infrastructural development, industrial piping, industrial warehousing, material handling system, industrial pollution control and environment management systems, water supply and sanitation, town planning, power projects, etc. CET represents a reservoir of technical & managerial expertise inherited over four decades of Indian Steel Industry. It has kept pace with changing times and made continuous efforts for updating skills of engineers through planned HRD programmes, collaborative arrangements with academia and other professional organisations of repute and acquiring up-to-date hardwares & softwares for engineering work. All of these are blended with a concern for clients' profitability to ensure that the clients get the most cost effective solution, tailor-made for their requirement.

Management Training Institute (MTI)

This apex training institute for management training in SAIL was set up in 1962 in Ranchi to fulfil the managerial development needs of senior executives of the company and thereby act as a catalyst for achieving organizational goals. It is one of the first management training centres to be set up in the corporate sector in India.

The Management Training Institute (MTI) assesses the training needs of senior executives, designs and executes need-based training programmes and disseminates modern management thinking through its publications. It is involved in preparing trainer manuals, case studies, exercises and business games. MTI

designs company-wide HRD interventions, organizes senior level management workshops, conducts problem solving workshops for middle level executives and also leadership interventions for Junior level executives. MTI also offers some selected programmes to executives of other organisations.

MTI, as a corporate institute, monitors the overall progress of training activities in SAIL. It conducts network meetings for selected and important programmes. E-abhigyan, the e-learning portal of SAIL has been developed and maintained by MTI with vast technical and managerial learning repositories and facility for online assessment and certification. E-abhigyan has helped in creating an anytime anywhere e-learning system for all employees of SAIL.

Environment Management Division (EMD)

Environment Management Division (EMD), established on 23rd June, 1988, is the corporate unit of Steel Authority of India Limited (SAIL) headquartered at Kolkata and is certified with Quality Management System (QMS) linked with ISO 9001: 2015. The division is under the Directorate of Technical, Project & Raw Materials and headed by the Executive Director (EMD).

The primary role of EMD is to apprise the management of environment and pollution control activities at plants, mines and units located across India. It also plays a pivotal role in consolidation of the efforts of the plants, mines and units towards environment protection and resource optimization through its multifarious activities. Overall activities of EMD are described below:



SAIL, as a responsible corporate organization, is always committed to maintain a clean and sustainable environment in and around its plants and units by integrating sound environmental practices for control and prevention of pollution from all its activities.

Growth Division (GD)

Growth Division (GD) functions as a nodal agency for manufacture and supply of various spare parts and equipment to the SAIL Plants by utilizing available in-house facilities and vendor base. GD functions focus on effective utilization of the engineering shops in the steel plants. Main objectives of GD are: –

- Effective utilization of captive engineering facilities of each steel plant.
- Providing technical help to manufacture specialised equipment to cater to present requirement as well as long-term expansion and modernisation.
- Undertake projects within SAIL plants or outside.

SAIL Digital Transformation Division (SDTD)

SAIL Digital Transformation Division is the nodal agency for Digital & AI Interventions in SAIL. Under the aegis of SDTD, PRAVARTANAM is a 3-year Digital Transformation initiative launched by SAIL, designed to establish a seamlessly Connected Digital Enterprise. This program will harness cutting-edge technologies such as Artificial Intelligence, cloud computing, and advanced data analytics to revolutionize operations across SAIL's five Integrated Steel Plants, mining operations, and critical functions including logistics, sales and marketing, and procurement. With a vision to establish SAIL as an industry leader in digital innovation, PRAVARTANAM targets to deliver an annual value realization of ₹1,000 crore. Additionally, the initiative aims to cultivate a self-sustaining digital ecosystem that fuels long-term growth, operational excellence, and sustained competitiveness in the global steel industry.

SAIL's Digital Transformation Program will be rolled out in a structured, two-phase approach. Phase I will focus on formulating a robust digital roadmap to guide SAIL's transition from its current "As-Is" state to a globally benchmarked "To-Be" vision. Phase II will center on the implementation of prioritized digital use cases, with a strong focus on delivering measurable business value. This will be supported by change management efforts, extensive hands-on training, and technology deployment.

Directorates at Corporate Office and SAIL Board

Directorates at Corporate Office (CO)

The main function of the Corporate Office is to integrate the functioning of Plants/Units to improve synergy of the total operations of SAIL. This is achieved through various Directorates of the Corporate Office as under:

Mines Directorate

Finance Directorate

Personnel Directorate

Commercial Directorate

The Corporate Office is an overall policy-making body responsible for providing all the necessary help and support to the units for implementing the policies of the Company by coordinating with the various organizations and Government Departments.

SAIL Board :

The SAIL Board comprises of Four Functional Directors, two Government Directors & two Independent Directors and is headed by Chairman, SAIL. It enables SAIL to perform the following general functions at the corporate level.

- Long term strategic planning for the Company
- Policy formulation in consultation with plant personnel. Getting agreed action plans for implementation of the policies and ensuring their fulfillment. Achieving clarity and organizational commitment on objectives, goals and plans of action.
- Developing norms of performance in every functional area and ensuring commitment of progressively improved norms.
- Ensuring smooth and efficient operations and achievement of optimal performance of existing resources. Ensuring fulfillment of targets and orderly growth of the Company. Organizational development to maximize efficiency of the company.
- Reviewing performance of each unit with respect to target and suggesting corrective action where necessary.
- Achievement of well-coordinated functioning of different plants: improving inter-plant interactions, dissemination of knowledge and achieving synergy in Company's operations.
- Centralized control of Finance, Sales, Purchase/ Import of inputs.
- Capital investment decisions beyond power delegated to plant CEOs.
- Coordination with all external agencies, Central and State Governments Ministries, Railways, suppliers etc. in order to improve overall company operations.
- Development of an efficient and well-designed data bank and MIS at all levels within the organization to assist in problem identification and resolution.
- Projection of corporate image of the Company through media to the public in general.
